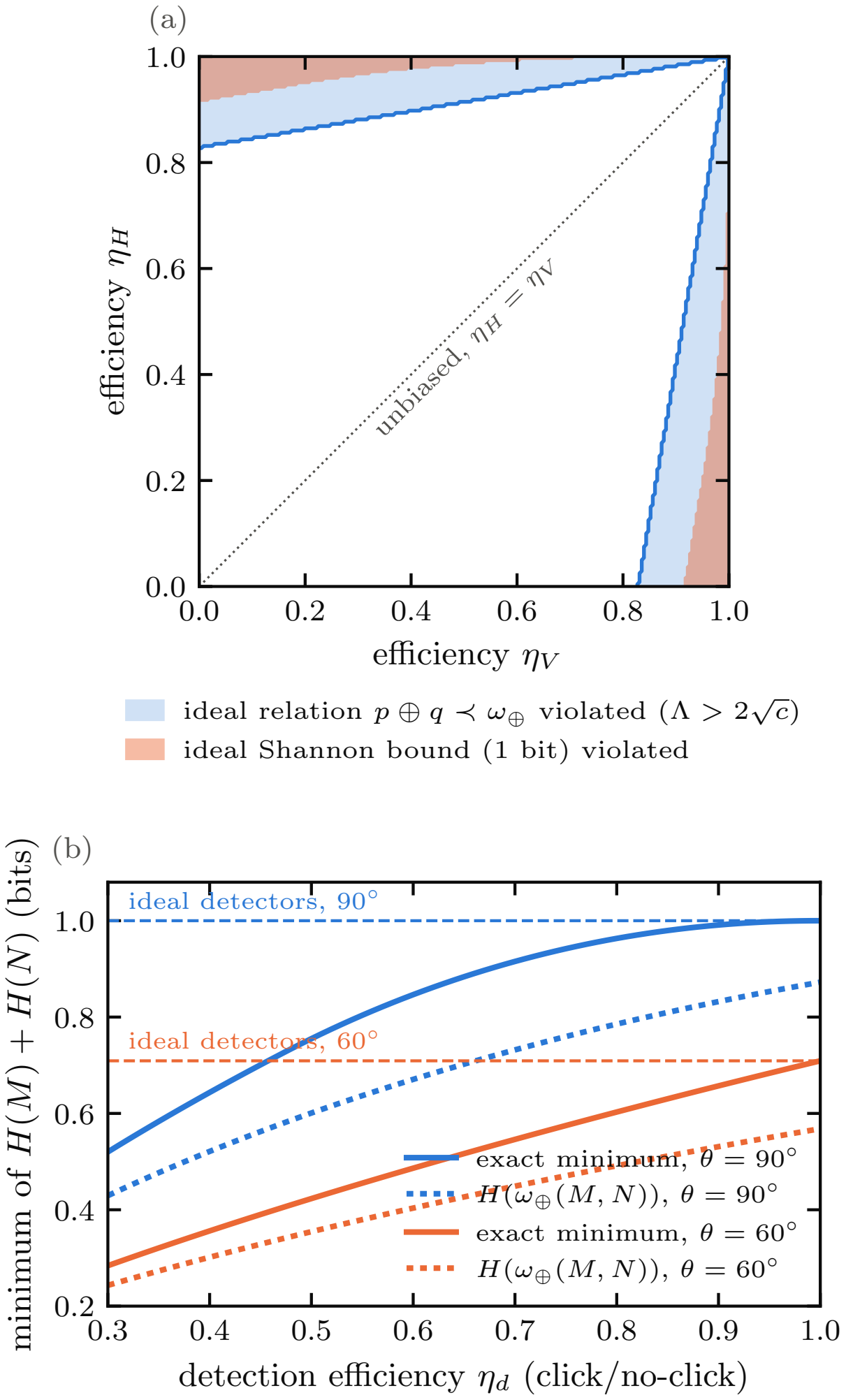

(a)
efficiency η_H
efficiency η_V
unbiased, $\eta_H = \eta_V$
ideal relation $p \oplus q \prec \omega_\oplus$ violated ($\Lambda > 2\sqrt{c}$)
ideal Shannon bound (1 bit) violated
(b)
minimum of $H(M) + H(N)$ (bits)
detection efficiency η_d (click/no-click)
ideal detectors, 90°
ideal detectors, 60°
exact minimum, $\theta = 90°$
$H(\omega_\oplus(M, N))$, $\theta = 90°$
exact minimum, $\theta = 60°$
$H(\omega_\oplus(M, N))$, $\theta = 60°$